

COMPARATIVE ANALYSIS

1. Object Detection Performance Comparison

The two object detection models, Viola-Jones and YOLO, are compared using True Positives (TP), False Positives (FP), and False Negatives (FN).

Model	TP	FP	FN
Model A – Viola-Jones	80	20	30
Model B – YOLO	90	15	20

(a) Calculation of Precision and Recall

Precision measures how many of the detected objects are actually correct. Recall measures how many of the actual objects are successfully detected.

Formula:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Model A – Viola-Jones

$$\text{Precision} = 80 / (80 + 20) = 80 / 100 = 0.80 = 80\%$$

$$\text{Recall} = 80 / (80 + 30) = 80 / 110 = 0.7273 = 72.73\%$$

Model B – YOLO

$$\text{Precision} = 90 / (90 + 15) = 90 / 105 = 0.8571 = 85.71\%$$

$$\text{Recall} = 90 / (90 + 20) = 90 / 110 = 0.8182 = 81.82\%$$

Model	Precision	Recall
Viola-Jones	80%	72.73%
YOLO	85.71%	81.82%

(b) Comparison

YOLO has higher precision and higher recall than Viola-Jones. Its precision is 85.71%, compared with 80% for Viola-Jones. Its recall is 81.82%, compared with 72.73% for Viola-Jones.

YOLO also has fewer false positives and false negatives. Therefore, it is better at correctly identifying objects and at detecting a larger proportion of the actual objects.

(c) Justification

YOLO provides better overall object detection performance because it achieves higher precision and recall. This indicates fewer incorrect detections and fewer missed objects.

Practical significance: For surveillance, autonomous vehicles, and real-time monitoring, high precision and recall are important because incorrect detections and missed objects can reduce system reliability.

Final Answer: YOLO provides better overall object detection performance.

2. Motion Estimation Error Analysis

The displacement errors for three consecutive frames are:

Method	Frame 1	Frame 2	Frame 3
Method A	2 pixels	3 pixels	2 pixels
Method B	3 pixels	5 pixels	4 pixels

(a) Average Displacement Error

Formula:

Average Displacement Error = Sum of displacement errors / Number of frames

Method A:

$$\text{Average} = (2 + 3 + 2) / 3 = 7 / 3 = 2.33 \text{ pixels}$$

Method B:

$$\text{Average} = (3 + 5 + 4) / 3 = 12 / 3 = 4 \text{ pixels}$$

Method	Errors	Average Error
Method A	2, 3, 2	2.33 pixels
Method B	3, 5, 4	4 pixels

(b) Comparison

Method A has an average displacement error of 2.33 pixels, while Method B has an average error of 4 pixels. Since a lower displacement error indicates greater accuracy, Method A performs better.

Method A also shows smaller errors in all three frames, whereas Method B produces larger errors in every frame.

(c) Justification

Method A is more accurate because its average displacement error is lower:

$$2.33 \text{ pixels} < 4 \text{ pixels}$$

Therefore, Method A provides more accurate motion estimation. Lower and more consistent errors can improve the reliability of applications such as object tracking, video analysis, surveillance, and motion-based computer vision systems.

Final Answer: Method A provides more accurate motion estimation.

3. Depth Estimation Comparison

The stereo vision and Structure-from-Motion (SfM) systems estimate the depth of three objects.

The actual depths are used to calculate the absolute error.

Object	Actual Depth	Stereo Vision	SfM
1	2.0 m	2.0 m	2.5 m
2	3.0 m	3.0 m	3.8 m
3	4.0 m	4.0 m	5.0 m

(a) Absolute Error for Each Object

Formula:

$$\text{Absolute Error} = |\text{Estimated Depth} - \text{Actual Depth}|$$

Object 1:

$$\text{Stereo Vision} = |2.0 - 2.0| = 0 \text{ m}$$

$$\text{SfM} = |2.5 - 2.0| = 0.5 \text{ m}$$

Object 2:

$$\text{Stereo Vision} = |3.0 - 3.0| = 0 \text{ m}$$

$$\text{SfM} = |3.8 - 3.0| = 0.8 \text{ m}$$

Object 3:

$$\text{Stereo Vision} = |4.0 - 4.0| = 0 \text{ m}$$

$$\text{SfM} = |5.0 - 4.0| = 1.0 \text{ m}$$

Object	Stereo Vision Error	SfM Error
1	0 m	0.5 m
2	0 m	0.8 m
3	0 m	1.0 m

(b) Average Absolute Error

Stereo Vision:

$$\text{Average} = (0 + 0 + 0) / 3 = 0 \text{ m}$$

SfM:

$$\text{Average} = (0.5 + 0.8 + 1.0) / 3 = 2.3 / 3 = 0.767 \text{ m}$$

Method	Average Absolute Error
Stereo Vision	0 m

SfM	0.767 m
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(c) Comparison and Justification

Stereo Vision provides an average absolute error of 0 m, while SfM has an average error of approximately 0.767 m. Lower error indicates more accurate depth estimation.

Stereo Vision gives estimated depths that exactly match the actual depths for all three objects.

Therefore, based on the given data, Stereo Vision provides more accurate depth estimation than SfM.

Practical significance: Accurate depth estimation is important in autonomous vehicles, robotics, 3D reconstruction, augmented reality, and navigation.

Final Answer: Stereo Vision provides more accurate depth estimation.

Overall Comparative Conclusion

The three questions demonstrate that model selection should be based on appropriate performance measures rather than simply choosing a method without analysis.

1. Object Detection: YOLO performs better than Viola-Jones because it has higher precision and recall.
2. Motion Estimation: Method A is more accurate because it has a lower average displacement error of 2.33 pixels.
3. Depth Estimation: Stereo Vision is more accurate because it has zero average absolute error.

Overall, the selected methods are:

YOLO → Best Object Detection

Method A → Best Motion Estimation

Stereo Vision → Best Depth Estimation

These conclusions are supported by the calculated numerical results and the comparison of errors and detection measures.